Prognostic Value of Frailty in Chronic Limb-Threatening Ischemia (CLTI) Revascularization: A Systematic Review and Meta-Analysis

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Background: Frailty is an important predictor of adverse outcomes in vascular surgery, but its prognostic value after lower extremity revascularization for peripheral artery disease (PAD) remains uncertain. This systematic review and meta-analysis evaluated the association between frailty and outcomes after PAD revascularization.

Methods: PubMed, Embase, and Cochrane Library were searched from inception to May 12, 2026. Studies assessing frailty in patients undergoing peripheral artery surgery (endovascular, open, or hybrid lower extremity revascularization) for PAD were included. Random-effects meta-analyses were performed using odds ratios (ORs) or hazard ratios (HRs) with 95% confidence intervals (CIs). Prespecified subgroup analyses were conducted according to frailty assessment approach (frailty index-type vs. nonfrailty index tools) and procedural strategy.

Results: Nineteen studies comprising 1,638,543 patients were included. Because some studies enrolled mixed PAD populations, only patients undergoing lower extremity revascularization contributed to the pooled analyses. Frailty was associated with increased early mortality (OR: 4.74; 95% CI: 1.75−12.87), late mortality (HR: 1.59; 95% CI: 1.07−2.36), and major adverse cardiovascular events (OR: 2.56; 95% CI: 1.15−5.71), but not with major amputation, major adverse limb events, or postoperative complications. Subgroup analyses demonstrated no significant differences according to frailty assessment approach or procedural strategy.

Conclusion: Frailty is associated with substantially worse outcomes following lower extremity revascularization for PAD, particularly increased early and late mortality and major adverse cardiovascular events, consistently across different frailty assessment approaches. These findings support routine incorporation of frailty assessment into preoperative evaluation to improve risk stratification, shared decision-making, and perioperative optimization.

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INTRODUCTION

Peripheral artery disease (PAD) affects approximately 200 million people globally and represents a major cause of functional impairment, cardiovascular morbidity, and limb loss. In patients with lifestyle-limiting claudication or chronic limb-threatening ischemia (CLTI), revascularization, whether endovascular, open, or hybrid, remains a fundamental therapeutic strategy to improve perfusion, preserve limb viability, and reduce symptoms.[1] However, patients undergoing PAD revascularization are frequently older, multimorbid, and physiologically vulnerable, making accurate risk stratification essential for clinical decision-making.

Frailty has emerged as an important marker of vulnerability in vascular surgery, distinct from chronological age alone.[2–4] It reflects diminished physiological reserve and reduced tolerance to procedural stress, which may influence perioperative recovery, long-term survival, functional outcomes, and discharge disposition. This is particularly relevant in PAD, where treatment decisions often require balancing limb salvage, procedural risk, life expectancy, and patient-centered goals of care. Notably, frailty is captured by a range of instruments, deficit-accumulation indices, phenotype-based scales, and administrative risk scores, that assess overlapping but nonidentical constructs, which complicates comparison of prognostic estimates across studies.

The burden of frailty in PAD is substantial, with an estimated prevalence of 49% in the general PAD population and 54% among patients with CLTI.[5] Prior meta-analyses have shown that frailty is associated with increased 30-day and long-term mortality after revascularization.[6] Consistently, recent guidelines recognize frailty as highly predictive of 30-day mortality across PAD revascularization procedures.[7] Nevertheless, the magnitude and consistency of this association across different clinical outcomes and revascularization strategies remain incompletely defined.

Three limitations in the available evidence justify a PAD-specific synthesis. First, frailty assessment remains heterogeneous, with limited agreement between commonly used instruments, including frailty indices, clinical frailty scales, and functional measures.[3,4] Second, it remains uncertain whether the prognostic impact of frailty differs according to the surgical approach, particularly endovascular versus open surgical treatment or primary amputation.[2] Third, previous syntheses have often included heterogeneous vascular surgery populations or focused on isolated outcomes, limiting their applicability to patients undergoing revascularization specifically for PAD.[8,9]

Beyond incorporating more recent publications, the present review advances prior syntheses by restricting pooled analyses to patients undergoing lower extremity revascularization specifically for PAD, applying a three-level meta-analytic model to account for nonindependent effect sizes derived from the same cohorts, and performing an exploratory meta-regression to investigate patient age as a potential source of heterogeneity. A clearer estimate of the risk associated with frailty in this population may support shared decision-making, improve preoperative counseling, guide optimization strategies, and help identify patients who may benefit from multidisciplinary assessment before intervention.[1,2] Although current guidelines acknowledge frailty as an important consideration, specific recommendations regarding its systematic assessment in PAD revascularization remain limited.[1] Therefore, this systematic review and meta-analysis aimed to synthesize the available evidence comparing clinical outcomes between frail and nonfrail patients undergoing revascularization for PAD.

METHODS

Protocol and Registration

We performed the systematic review and meta-analysis according to the Cochrane Handbook for Systematic Reviews of Interventions and structured it according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA 2020) guidelines.[10,11] The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD420261393990.[12] No separate protocol document was published. No amendments were made to the registered protocol during the conduct of the review.

Eligibility Criteria

Studies were included if they met any of the following criteria: (1) patients with PAD presenting with chronic limb-threatening ischemia (CLTI) undergoing peripheral artery surgery, defined throughout this review as endovascular, open, or hybrid lower extremity revascularization for PAD (including major amputation when reported as a treatment arm); (2) assessment of preoperative frailty using validated instruments, including judgment-based or phenotype-like measures such

as the Clinical Frailty Scale (CFS), Rockwood Clinical Frailty Scale, Risk Analysis Index (RAI), Vascular Quality Initiative-derived RAI (VQI-RAI), and Barthel Index, as well as deficit accumulation/frailty index–type measures such as the frailty index (FI), modified frailty index (mFI-5 and mFI-11), simplified frailty index (sFI), hospital frailty risk score (HFRS), VQI Frailty Score (VQI-FS), CLI Frailty Index, and CLTI Frailty Risk Score; (3) separation of outcomes by frailty and nonfrailty status; and (4) reporting of at least one of the outcomes of interest. Studies were excluded if they met any of the following criteria: (1) failure to specify the vascular segment involved; (2) overlapping study populations; (3) unavailable full text; or (4) publication types including case reports, conference abstracts, meta-analyses, review articles, or animal studies.

Search Strategy and Study Selection

We systematically searched PubMed, Embase, and Cochrane Library databases from inception to May 12, 2026. The search strategy for all databases was: (frailty OR frail OR "frailty index" OR "clinical frailty scale" OR mFI OR "modified frailty index") AND ("peripheral artery disease" OR "peripheral arterial disease" OR PAD OR "peripheral vascular disease" OR PVD OR "lower extremity arterial disease" OR LEAD OR "chronic limb-threatening ischemia" OR CLTI OR "critical limb ischemia" OR CLI OR "limb ischemia" OR "lower limb ischemia"). We also searched the references of the included studies and previous systematic reviews and meta-analyses aiming for the inclusion of additional studies.[13]

Two authors (M.L.C. and I.O.T.) independently conducted the search, imported results into Rayyan, a web-based systematic review tool, and triaged the studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the review of the full-text articles. Disagreements were solved by consensus.

Outcomes and Subgroup Analysis

The outcomes of interest were as follows: (1) early mortality; (2) late mortality; (3) major postoperative complications, accepted according to each study's own definition (e.g., wound infection, cardiopulmonary complications, or reoperation) in the absence of a uniform definition across studies; (4) major adverse cardiovascular events (MACE), including cardiovascular death, myocardial infarction, and stroke according to individual study definitions; (5) amputation; (6) major adverse limb events (MALE), when reported, included major amputation and limb-related reinterventions according to study-specific definitions. We performed subgroup analyses to explore potential sources of heterogeneity. These analyses were stratified according to the frailty assessment tool (frailty index-type versus judgment-based/phenotype-like), surgery type (endovascular versus open). Instruments were grouped into Frailty Index-type (deficit-accumulation) versus judgment-based/phenotype-like categories because, despite assessing related but not identical constructs, both approaches share the aim of quantifying cumulative physiological vulnerability and are the two broad classification schemes most consistently used in the vascular surgery frailty literature, allowing comparability with prior syntheses. In addition, we further specified the frailty index-type scale (mFI-5 versus mFI-11).

When a study reported more than one eligible effect estimate for the same outcome domain (e.g., individual MACE components such as myocardial infarction, stroke, and cardiovascular death), all reported estimates were extracted, and the correlation among these estimates within the same cohort was subsequently accounted for using the three-level meta-analytic model described below.

Data Extraction

Two authors (M.L.C. and I.O.T.) independently extracted data from the included studies using a standardized data extraction form. The following information was collected: (1) general study characteristics, including first author, year of publication, country, study design, and study period; (2) patient and clinical characteristics, including total number of patients, mean or median age, and sex distribution; (3) type of procedure; (4) frailty-related variables, including frailty assessment tool, frailty cutoff definition, proportion of frail patients, and duration of follow-up; and (5) reported outcomes. For studies enrolling mixed PAD populations (e.g., cohorts combining revascularization, primary amputation, and conservative management), outcome data specific to the lower extremity revascularization subgroup were extracted directly from subgroup-stratified results reported in the original tables or supplementary material; studies that did not report revascularization-specific outcomes separately were not included in the relevant pooled analysis. Studies requiring such subgroup-specific extraction are noted in the study characteristics table.

Statistical Analysis

Treatment effects for binary outcomes were pooled using odds ratios (ORs) and time-to-event outcomes using hazard ratios (HRs), both with corresponding 95% confidence intervals (CIs). For each study, the log-transformed effect estimate (logOR or logHR) and its standard error were used; when the standard error was not directly reported, it was derived from the reported 95% confidence interval as (log(upper limit) - log(lower limit))/3.92. Between-study heterogeneity was assessed using Cochran's Q test and the I^2 statistic. *P* values < 0.10 and I^2 values of 25%, 50%, and 75% were interpreted as indicating low, moderate, and high heterogeneity, respectively. Random-effects models were fitted using the restricted maximum likelihood estimator. For the primary meta-analyses, 95% CIs were calculated using the Hartung–Knapp–Sidik–Jonkman method. For subgroup analyses, pooled estimates and 95% CIs were calculated using the Wald method, and subgroup differences were assessed using the chi-square test for interaction, with $P < 0.10$ considered statistically significant. All statistical analyses were performed using R (version 4.4.3; R Foundation for Statistical Computing, Vienna, Austria) using the meta package. Because two studies (Ali et al[14] and Balasundaram et al[15]) reported multiple correlated effect sizes for distinct MACE components derived from the same patient cohort, we additionally performed a three-level meta-analytic model (random = ~1|study, specified via the cluster argument in metagen) to account for the nonindependence of these estimates in the MACE subgroup analysis. To explore potential sources of the observed heterogeneity, we additionally performed a random-effects meta-regression for the early mortality outcome using study-level mean patient age as a continuous moderator, restricted to studies reporting age (metareg function, meta package).

Sensitivity Analysis

For outcomes with substantial heterogeneity ($I^2 \geq 25\%$), we performed leave-one-out sensitivity analyses to assess the robustness of the pooled estimates and explore potential sources of heterogeneity. In this approach, each study was sequentially omitted, and the meta-analysis was repeated to evaluate the influence of individual studies on the overall effect estimate and between-study heterogeneity. This analysis provides insight into the stability of the overall estimates and whether individual studies exert undue influence on the results.[16]

Risk of Bias Assessment

The risk of bias of each included study was assessed with the Risk of Bias in Nonrandomized Studies of Interventions tool (ROBINS-I).[17] Two independent authors conducted the risk of bias assessment (M.L.C. and L.M.D.) and disagreements were solved by consensus. Publication bias was assessed when at least 10 studies reported the same outcome, using visual inspection of contour-enhanced funnel plots and formal testing with Egger's regression asymmetry tests.[10,18]

Certainty of evidence for each outcome was not formally assessed using the GRADE framework, as the primary aim of this review was to quantify the association between frailty and clinical outcomes rather than to inform a specific clinical recommendation. Given the extreme between-study heterogeneity and predominance of retrospective observational designs described below, the overall certainty of evidence for the pooled estimates is likely low to very low by GRADE criteria; this is acknowledged as a limitation, and formal certainty rating should be undertaken in future updates of this synthesis.

RESULTS

Study Selection

As detailed in Figure 1, the initial database search identified 9,532 records, including 2,709 from PubMed, 6,023 from Embase, and 800 from Cochrane. After the removal of 2,058 duplicate records, 7,474 records were screened by title and abstract. Of these, 7,343 records were excluded, and 131 reports were sought for retrieval and assessed for eligibility. No reports were unavailable for retrieval. After full-text assessment, 112 reports were excluded for the following reasons: abstract only ($n = 37$), no outcomes of interest ($n = 24$), no direct frail versus nonfrail comparison ($n = 22$), outcomes not reported by vascular segment/PAD subgroup ($n = 15$), full-text unobtainable ($n = 4$), overlapping population ($n = 3$), no population of interest ($n = 2$), review/editorial/protocol ($n = 4$), and not CLTI ($n = 1$). Ultimately, 19 studies were included in the systematic review. Supplementary Table I lists the reports excluded during full-text screening and the reasons for exclusion.

Study-Level and Patient-Level Characteristics

A total of 19 studies, published between 2016 and 2026, were included in this systematic review and

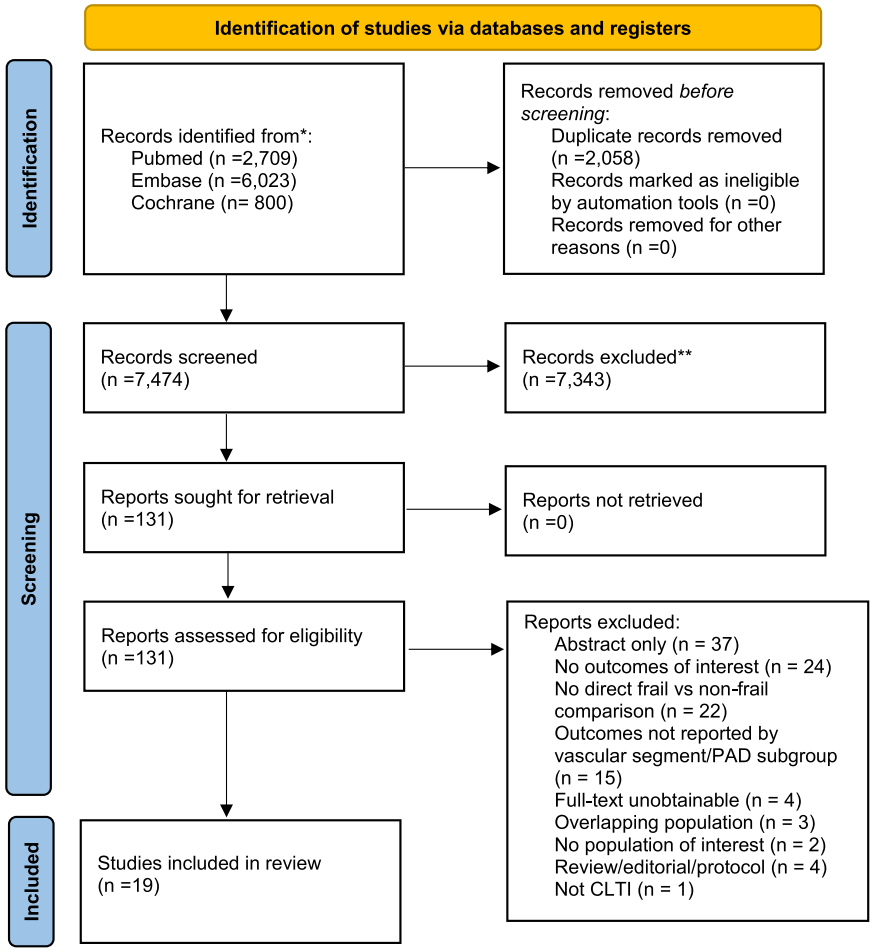


Fig. 1. PRISMA flow diagram. Study identification, screening, eligibility assessment, and inclusion process.

meta-analysis. Three studies were prospective, whereas the remaining studies had retrospective cohort designs. Overall, the included studies comprised 1,638,543 patients, with sample sizes ranging from 76 to 1,414,080 participants. Although the included studies enrolled patients managed with different treatment strategies, quantitative analyses were restricted to patients undergoing lower extremity revascularization for each outcome. Most studies were conducted in the United States, followed by the United Kingdom, Japan, and Singapore. The mean age of participants ranged from 63.6 to 93.0 years, and the proportion of male patients ranged from 36.0% to 100.0%. The included studies encompassed endovascular, open, and hybrid lower extremity revascularization procedures, whereas some also included patients undergoing major amputation or conservative management.

Frailty was assessed using heterogeneous instruments. The most frequently used tools were modified Frailty Index (mFI)-based measures (mFI, mFI-5, and mFI-11), followed by the CFS. Other assessment tools included the RAI, VQI-RAI, HFRS, sFI, Barthel Index, and the chronic limb-threatening ischemia frailty risk score (CLTI-FRS). Reported frailty prevalence varied widely across studies, ranging from 2.6% to 88.0%, reflecting differences in study populations, frailty definitions, and cutoff values. Follow-up ranged from in-hospital assessment to 24 months. Detailed study characteristics are presented in Table I.

Pooled Analyses

Frailty was associated with a significantly increased risk of early mortality following lower extremity revascularization. The pooled analysis demonstrated an OR of 4.74 (95% CI 1.75−12.87; $P = 0.0056$; $I^2 = 98.5\%$; Fig 2A). Likewise, frailty was associated with a significantly higher risk of

Table I. Characteristics of included studies

Study	Country	Period	Design	Data source	No. of patients	Procedure/population	Mean age, years	Male, %	Frailty assessment tool	Frailty cutoff	Frail, %	Follow-up, months
Ali et al[14]	United States	2010	Retrospective cohort	ACS-NSQIP	4,704	Infrainguinal lower extremity bypass surgery	67.9	64.0	mFI	>0.54	2.6	1
Andersen et al[19]	United States	2012–2017	Retrospective cohort	VQI registry	2,040	Below-knee amputation	63.6	68.0	mFI-5	Continuous	NR	1
Balasundaram et al[15]	United States	2015–2019	Retrospective cohort	NSQIP	11,947	Lower extremity endovascular revascularization	69.0	60.4	mFI-5	≥0.6	13.1	1
Bradley et al[20]	United Kingdom	2020–2021	Retrospective cohort	Single-center hospital registry	190	Mixed management; majority undergoing revascularization	NR	62.0	CFS	>4	53.0	22
Braet et al[21]	United States	2011–2017	Retrospective cohort	NSQIP	11,530	Lower extremity bypass surgery	NR	64.0	mFI-5	≥0.6	19.0	1
Brahmbhatt et al[22]	United States	2005–2012	Retrospective cohort	NSQIP	24,624	Infrainguinal vascular procedures, 92% open and 8% endovascular	67.9	63.9	mFI-11	≥0.25	60.0	1
Cotton et al[23]	United States	2014–2019	Retrospective cohort	Institutional registry	298	Above- or below-knee amputation	67.0	100.0	mFI-11	≥0.45	66.0	12
DeRieux et al[24]	United States	2009–2021	Retrospective cohort	Electronic medical record	76	Endovascular intervention in 87%	93.0	36.0	mFI-11	≥0.27	35.0	12
Gonzalez et al[25]	United States	2003–2016	Retrospective cohort	Single-center VA database	431	Open and endovascular lower limb revascularization	66.0	100.0	mFI-11	≥0.51	19.0	24
Gonzalez et al[26]	United States	2011–2020	Retrospective cohort	VQI registry	3,620	Open lower extremity bypass revascularization for acute limb ischemia	65.0	72.1	RAI	>20	78.0	12
Hart and Davies[27]	United States	2018–2022	Retrospective cohort	Institutional registry	1,859	Endovascular treatment, bypass, amputation, or wound care	65.0	56.0	VQI-RAI	≥35	25.0	12
Houghton et al[28]	United Kingdom	2018–2019	Retrospective cohort	Single-center vascular limb salvage clinic	190	Mixed CLTI management, including endovascular, open, and conservative treatment	74.0	66.0	CFS	≥5	52.0	12
Houghton et al[29]	United Kingdom	2019–2022	Prospective cohort	Hospital-based cohort	363	CLTI treated with revascularization and/or medical management	75.0	NR	CFS	≥5	NR	12
Karim et al[30]	United States	2012–2015	Retrospective cohort	National Inpatient Sample	1,414,080	CLTI, mixed medical and revascularization management	68.0	61.0	CLTI-FRS	>0.40	23.5	In-hospital
Kodama et al[31]	Japan	2007–2016	Retrospective cohort	Single-center registry	107	Infrainguinal bypass for CLTI	69.0	65.0	Barthel Index	<70 vs > 75	33.6	24
Majmundar et al[32]	United States	2015–2018	Retrospective cohort	USRDS	122,649	PAD with or without revascularization	66.0	55.0	HFRS	≥5	NR	12
Pandit et al[33]	United States	2005–2012	Retrospective cohort	NSQIP	8,681	Above- and below-knee amputations	76.0	68.0	mFI-5/mFI-11	≥0.27	24.5–29.8	1
Schweitzer et al[34]	United States	2014–2018	Retrospective cohort	NSQIP	30,921	Lower extremity bypass surgery	67.0	58.0	sFI	≥0.6	NR	1
Soon et al[35]	Singapore	2015–2017	Retrospective cohort	NR	233	Endovascular revascularization	69.0	58.8	mFI-11	≥3	88.0	12

ACS-NSQIP, American College of Surgeons National Surgical Quality Improvement Program; CLTI-FRS, chronic limb-threatening ischemia frailty risk score; GNRI, Geriatric Nutritional Risk Index; NR, not reported; SMI, skeletal muscle index; USRDS, United States Renal Data System; VA, veterans affairs.

For studies enrolling mixed PAD populations (see Procedure/population column, e.g., Bradley 2024, Houghton 2021, Houghton 2026, Karim 2022, and Majmundar 2025), outcome data specific to the lower extremity revascularization subgroup were extracted separately from subgroup-stratified results reported in the original study, as detailed in the Methods.

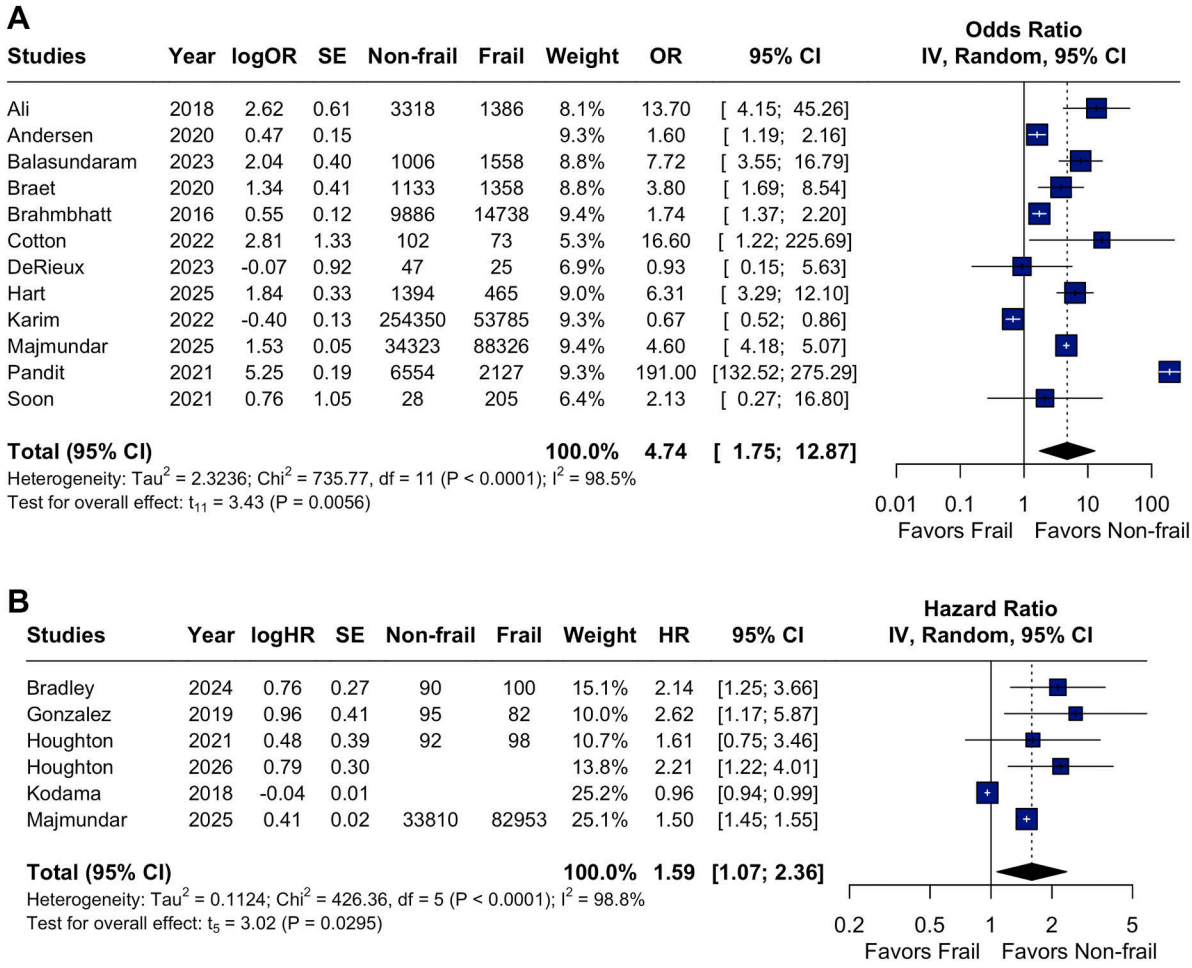


Fig. 2. Frailty and mortality after lower extremity revascularization. Forest plots for the association between frailty and (**A**) early mortality, and (**B**) late mortality.

late mortality, with a pooled HR of 1.59 (95% CI 1.07−2.36; $P = 0.0295$; $I^2 = 98.8\%$; Fig 2B).

Regarding postoperative morbidity, frail patients had significantly higher odds of MACE (OR: 2.56; 95% CI: 1.15−5.71; $P = 0.0268$; $I^2 = 90.7\%$; Supplementary Fig S1). Although frailty was associated with increased odds of major amputation (OR: 1.69; 95% CI: 0.93−3.07; $P = 0.0770$; $I^2 = 81.3\%$; Supplementary Fig S2),MALE (OR: 2.33; 95% CI: 0.98−5.53; $P = 0.0530$; $I^2 = 97.8\%$; Supplementary Fig S3), and overall postoperative complications (OR: 1.44; 95% CI: 0.86−2.42; $P = 0.1514$; $I^2 = 96.1\%$; Supplementary Fig S4), these pooled estimates did not reach statistical significance. Between-study heterogeneity was consistently extreme across outcomes (I^2 81.3%−98.8%), and these pooled effect estimates should therefore be interpreted with caution.

Subgroup Analyses

Subgroup analysis demonstrated that the association between frailty and early mortality was not influenced by the type of frailty assessment tool ($P = 0.3813$; Fig 3). Frailty index-type instruments yielded a pooled OR of 5.90 (95% CI: 1.87 to 18.64), whereas judgment-based/phenotype-like instruments demonstrated a pooled OR of 2.65 (95% CI: 0.67 to 10.50). Although the point estimate was higher for frailty index-based tools, no significant subgroup difference was observed.

Subgroup analysis according to procedural strategy likewise demonstrated no significant subgroup differences ($P < 0.0001$ for heterogeneity between studies; subgroup difference test not significant). The association between frailty and early mortality remained significant among patients

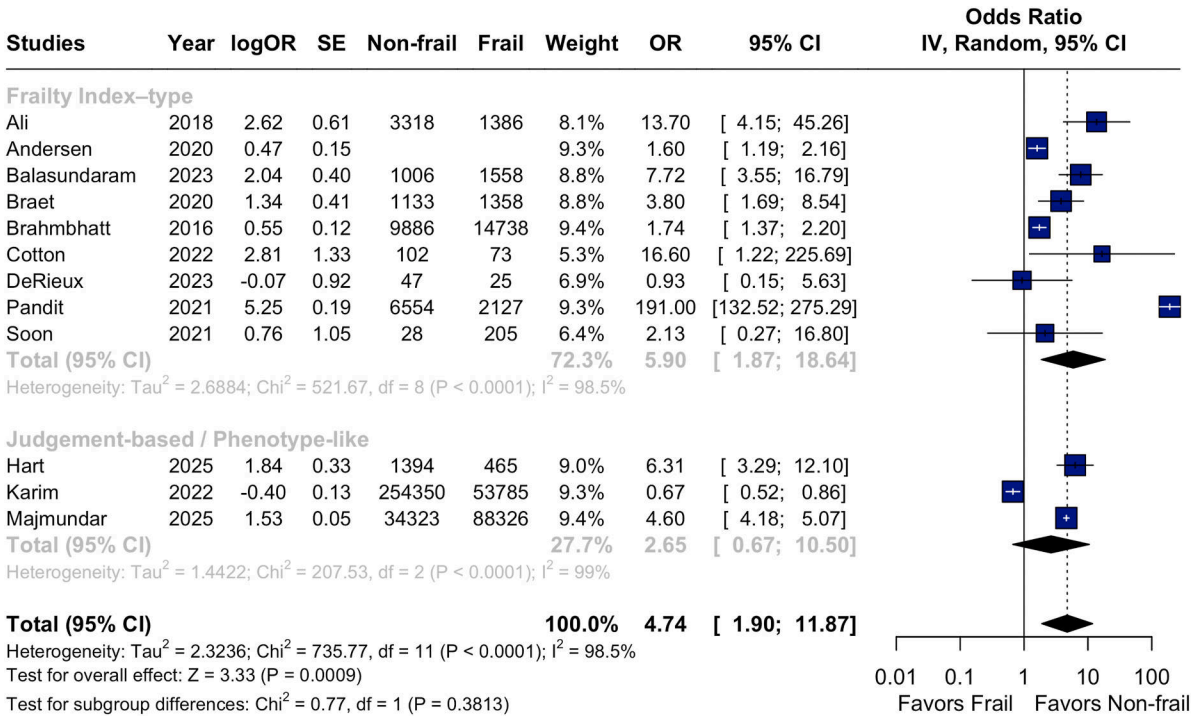


Fig. 3. Early mortality by frailty assessment tool. Subgroup analysis stratified by frailty index-type versus judgment-based/phenotype-like tools.

undergoing open revascularization (OR: 3.19; 95% CI: 1.38−8.43) and amputation (OR: 17.20; 95% CI: 1.92−321.44), whereas the association was not statistically significant in the endovascular subgroup (OR: 3.09; 95% CI: 0.67−14.14) or the mixed management subgroup (OR: 0.67; 95% CI: 0.52−0.86) (Supplementary Fig S5).

For MACE, because two studies (Ali et al. and Balasundaram et al.) contributed multiple correlated effect sizes derived from the same cohort, subgroup analysis according to frailty assessment approach was performed using a three-level meta-analytic model accounting for within-study correlation (cluster = study). Frailty index-type instruments yielded a pooled OR of 1.85 (95% CI 1.13 to 3.03), whereas non-frailty Index tools yielded a pooled OR of 3.20 (95% CI 0.56 to 18.36). Although the point estimate was higher for non-frailty index tools, a formal subgroup-difference test could not be computed under the three-level model; the point estimates were directionally consistent with the independent-effects model ($P = 0.5293$), suggesting no material subgroup difference. The overall pooled estimate remained significant (OR: 2.38; 95% CI: 1.25−4.54) **(**Supplementary Fig S6**).**

Similarly, subgroup analysis according to procedural strategy demonstrated no statistically significant subgroup differences ($P = 0.0550$). The association between frailty and MACE remained significant in open surgical procedures (OR: 1.53; 95% CI: 1.26−1.87), whereas the pooled estimate for endovascular procedures did not reach statistical significance (OR: 7.85; 95% CI: 1.36−55.40) because of the wide confidence interval and marked between-study heterogeneity **(**Supplementary Fig S7**).**

For major amputation, subgroup analysis demonstrated no significant differences according to frailty assessment tool ($P = 0.8407$). Frailty index-type tools yielded a pooled OR of 1.45 (95% CI: 0.89 to 2.39), whereas judgment-based/phenotype-like tools yielded an OR of 1.90 (95% CI: 0.75 to 4.87); neither subgroup reached statistical significance **(**Supplementary Fig S8**).**

Several of these subgroup estimates, particularly for MACE and amputation, were derived from as few as two to four studies per stratum; findings from these smaller strata should be regarded as exploratory and interpreted with caution rather than as definitive evidence of subgroup effect or equivalence.

Sensitivity Analyses

Leave-one-out analyses demonstrated that the pooled estimates remained directionally consistent across all outcomes. For early mortality, sequential exclusion of individual studies yielded pooled ORs ranging from 3.10 to 5.82, with persistently high heterogeneity (I^2 96.4%−98.6%) and no material change in the overall significance of the findings. For late mortality, pooled HRs ranged from 1.51 to 1.67, and exclusion of individual studies did not alter the statistical significance of the association, although omission of Kodama et al. produced the lowest heterogeneity. For MACE, exclusion of any single study did not substantially change the pooled estimate, with ORs ranging from 1.54 to 2.94. In contrast, for major amputation, exclusion of several individual studies produced attenuation of the pooled effect, with pooled ORs ranging from 1.32 to 1.89, and the overall association remained nonsignificant. Detailed leave-one-out analyses are presented in Supplementary Figures S9−S12.

To account for nonindependent effect sizes derived from the same cohort (Ali et al. and Balasundaram et al.), we additionally performed a three-level meta-analytic sensitivity analysis for the MACE subgroup comparison. The pooled OR for MACE decreased modestly from 2.56 (independent-effects model) to 2.38 (95% CI 1.25 to 4.54) after accounting for within-study correlation, and the association remained statistically significant ($P = 0.0082$), confirming that the primary finding is robust to violations of the independence assumption.

Risk of Bias and Quality Assessment

Overall, the risk of bias across the included studies was predominantly moderate to high, largely driven by concerns in the domains of performance bias and, to a lesser extent, selection and detection bias. Most studies demonstrated low risk for attrition and reporting bias, indicating generally complete outcome data and acceptable reporting practices. However, the observational design of the included cohorts, together with limited blinding and insufficient methodological detail, contributed to moderate or high risk ratings in several domains. The detailed risk of bias assessment is presented in Figure 4.

Visual inspection of the funnel plot revealed no marked asymmetry, and Egger's regression test showed no evidence of small-study effects ($t = 0.15$, df = 10, $P = 0.8837$; bias estimate = 0.5147, SE = 3.4308), suggesting no statistically significant evidence of publication bias for the early mortality analysis **(**Supplementary Fig S13**)**.

Meta-regression

To explore whether mean patient age influenced the magnitude of the frailty−early mortality association, a random-effects meta-regression was performed using study-level mean age as a continuous moderator (k = 11 studies with reported age). Age was not significantly associated with the log OR for early mortality ($\beta = 0.0010$; 95% CI − 0.1583 to 0.1603; $P = 0.9887$) and did not explain any of the observed between-study heterogeneity ($R^2 = 0.00\%$; residual $I^2 = 98.96\%$) **(**Supplementary Fig S14**)**.

DISCUSSION

This meta-analysis demonstrated that frailty is associated with worse outcomes in patients with lower extremity PAD undergoing revascularization. Overall, frail patients presented an approximately 4.7-fold higher risk of early mortality (OR: 4.74; 95% CI: 1.75−12.87) and a nearly 60% higher risk of late mortality (HR: 1.59; 95% CI: 1.07−2.36) following lower extremity revascularization. In addition, a robust association was observed between frailty and MACE (OR: 2.56; 95% CI: 1.15−5.71). Although amputation (OR: 1.69; 95% CI: 0.93−3.07) and MALE (OR: 2.33; 95% CI: 0.98−5.53) showed unfavorable trends among frail patients, these associations did not reach statistical significance, possibly due to the substantial heterogeneity observed across studies ($I^2 \geq 81\%$). The wide confidence interval around the early mortality estimate (95% CI: 1.75 to 12.87) likely reflects, rather than obscures, the considerable variability in frailty definitions, study populations, procedural strategies, and outcome ascertainment windows across the included cohorts; consequently, the pooled point estimate should be regarded as an approximate signal of association rather than a precise effect size. Given this extreme and largely unexplained heterogeneity, these findings should be interpreted as reinforcing frailty as a relevant prognostic marker in vascular surgery, whereas the magnitude of the pooled estimates warrants cautious, rather than definitive, clinical interpretation.

Previous observational studies support the association between frailty and adverse outcomes in PAD. In a cohort of more than 85,000 patients with CLTI, frailty, measured using a claims-based frailty indicator, was associated with a higher risk of amputation or death regardless of revascularization strategy.[36] Interestingly, frail patients were less likely to undergo surgical rather than endovascular revascularization, suggesting treatment selection based on

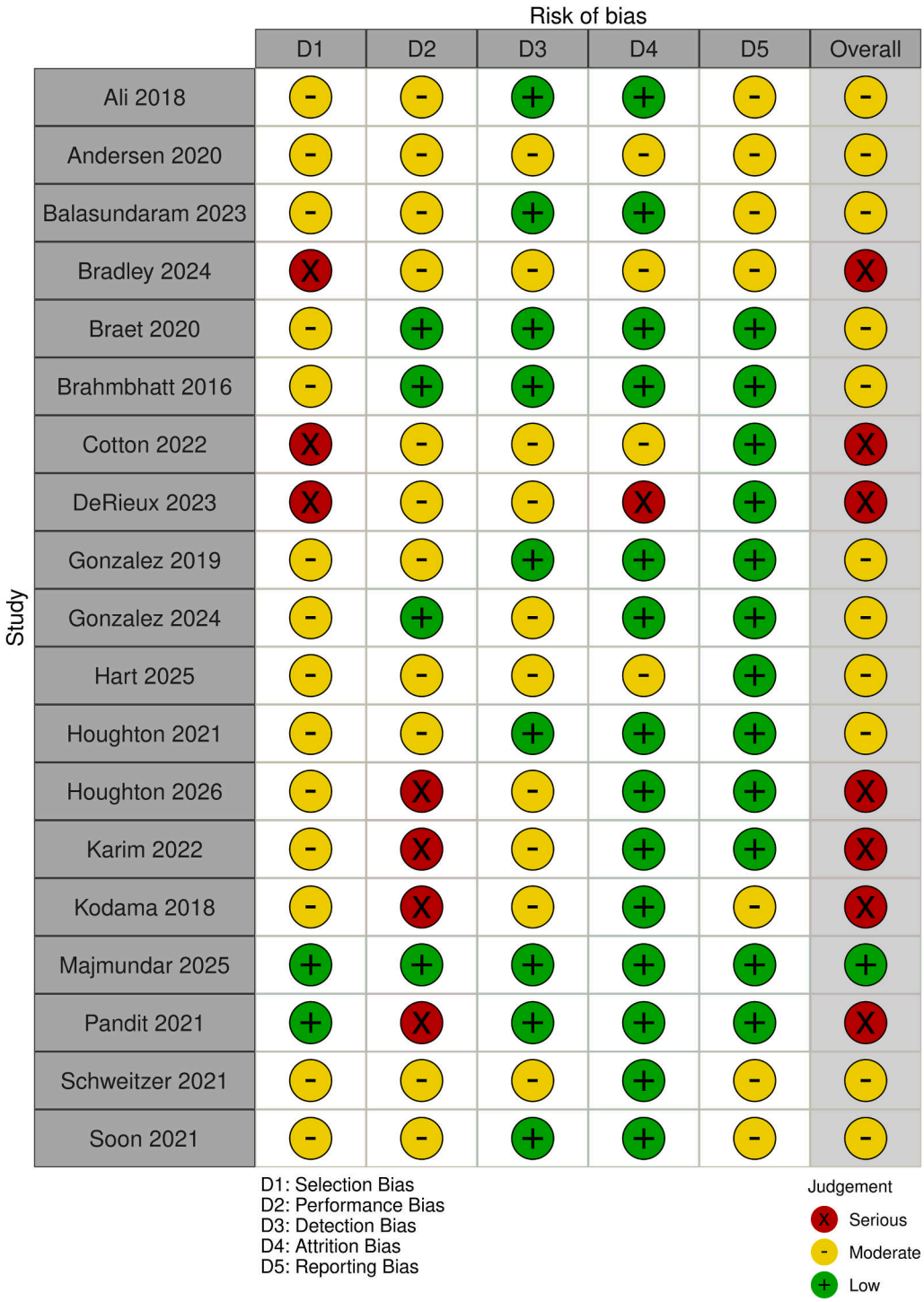


Fig. 4. Risk of bias assessment. ROBINS-I assessment across included studies.

functional status.[36] This pattern of treatment-selection (confounding by indication) is a plausible contributor to the associations observed in our pooled analyses as well, since frail patients are frequently directed toward less invasive strategies or excluded from revascularization altogether; such selection may attenuate or distort comparisons of outcomes by frailty status across procedural

subgroups and cannot be fully accounted for using aggregate, study-level data. The broader cardiovascular frailty literature demonstrates that frail individuals have a 1.8-fold higher risk of developing peripheral vascular disease and a 1.6−2.6-fold greater risk of cardiovascular mortality.[37−39] These data support the clinical relevance of our findings.

These findings carry important clinical implications. Systematic frailty assessment using instruments such as the mFI-5, Hospital Frailty Risk Score, or frailty index should be integrated into preoperative evaluation.[1,40] Population-based studies suggest that frail patients have a higher probability of death or amputation independent of revascularization strategy, indicating that frailty provides prognostic information beyond traditional surgical risk factors.[3,36] Although endovascular procedures are generally considered less invasive, our subgroup analyses demonstrated no significant differences in the prognostic impact of frailty according to revascularization strategy. Frailty remained associated with increased early mortality across procedural subgroups, with the strongest association observed among patients undergoing open revascularization and major amputation, whereas no statistically significant association was observed in the endovascular subgroup.[2,3,36,41] Similarly, subgroup analyses according to frailty assessment approach demonstrated no significant differences between frailty index-type and non-frailty index tools for either early mortality or MACE, suggesting that the direction of the prognostic association is broadly similar across different categories of validated frailty instruments. However, the absence of a statistically significant subgroup difference does not constitute evidence of equivalence between instruments, particularly given the limited number of studies contributing to several strata, and should not be interpreted as license to use these tools interchangeably. These findings nevertheless support the clinical utility of frailty assessment using any validated instrument, and suggest that the prognostic impact of frailty may be relevant across procedural strategies. Rather than using frailty status alone to determine treatment strategy, these findings support incorporating frailty assessment into shared decision-making, perioperative optimization, and prehabilitation planning, alongside anatomical complexity, overall physiologic reserve, and functional status.[1,9] Prehabilitation strategies, including supervised exercise therapy, nutritional optimization, and multidisciplinary perioperative management, may help mitigate the adverse effects of frailty.[1]

The present meta-analysis has several limitations. In summary, the principal limitations are substantial and largely unexplained between-study heterogeneity; heterogeneous and only partially overlapping frailty definitions and assessment instruments; variable and often incomplete adjustment for confounding across the included studies; and a predominance of retrospective observational cohort designs; each is discussed in turn below. The most important limitation is the substantial heterogeneity observed across several outcomes (I^2 ranging from 81.3% to 98.8%), likely reflecting differences in frailty assessment methods, outcome definitions, patient populations, and procedural strategies. Although subgroup analyses according to frailty assessment approach and procedural strategy were performed, substantial residual heterogeneity persisted, indicating that additional unmeasured clinical and methodological factors probably contributed to the observed variability. In addition, most included studies were retrospective observational cohorts, making them susceptible to selection bias, lack of blinding, and residual confounding, which reduces the certainty of the available evidence. Methodological quality was predominantly moderate to high risk of bias, particularly in the domains of performance and selection bias. Although visual inspection of the funnel plot and Egger's regression test did not demonstrate statistically significant evidence of publication bias, the relatively small number of studies available for some outcomes limited the power to detect small-study effects. Furthermore, the subgroup analyses were based on broad categories of frailty assessment tools, and some categories included only a small number of studies, limiting statistical power to detect subgroup differences. Future studies should standardize frailty assessment tools, apply more rigorous adjustment for comorbidities, and explore meta-regression approaches to better explain heterogeneity.

In conclusion, frailty is associated with worse prognosis in patients with PAD undergoing lower extremity revascularization, including higher early and late mortality and higher rates of MACE, although the extreme and largely unexplained between-study heterogeneity, heterogeneous frailty definitions, and predominantly retrospective study designs warrant cautious interpretation of the magnitude of these associations. Although the associations with amputation and MALE did not reach statistical significance, the direction of the observed effects suggests potential clinical relevance that merits further study. The available evidence supports consideration of frailty assessment as part of comprehensive preoperative evaluation, alongside anatomical and procedural factors, to inform shared decision-making and perioperative optimization. The absence of significant subgroup differences

across frailty assessment tools should not be interpreted as evidence that these instruments are interchangeable. Prospective studies using standardized, validated frailty instruments and reporting revascularization-specific outcomes remain necessary to clarify the magnitude of these associations before stronger clinical recommendations can be made.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

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SUPPLEMENTARY DATA

Supplementary data related to this article can be found at https://doi.org/10.1016/j.avsg.2026.08.048.

REFERENCES

1. Writing Committee MembersGornik HL, Aronow HD, Goodney PP, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on clinical practice guidelines. J Am Coll Cardiol 2024;83:2497—604.
2. Sorondo SM, Arnow K, Kashikar A, et al. Endovascular surgery does not mitigate the adverse effects of frailty on the risk of losing life or limb. J Vasc Surg 2025;82:1835—44.
3. Al-Damluji MS, Smolderen KG, Meng C, et al. Frailty and outcomes following revascularization of lower-extremity peripheral artery disease: insights from the Vascular Quality Initiative (VQI). Vasc Med 2022;27:251—7.
4. Drudi LM, Ades M, Mancini R, et al. Frailty assessment in older adults undergoing interventions for peripheral arterial disease. J Vasc Surg 2019;70:1594—1602.e1.
5. Wang Y, Wu X, Hu X, et al. Prevalence of frailty in patients with lower extremity peripheral arterial disease: a systematic review and meta-analysis. Ageing Res Rev 2022;82:101748.
6. Zhang H, Jie Y, Wang P, et al. Impact of frailty on all-cause mortality or major amputation in patients with lower extremity peripheral artery disease: a meta-analysis. Ageing Res Rev 2022;79:101656.
7. Writing Committee MembersGornik HL, Aronow HD, Goodney PP, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on clinical practice guidelines. J Am Coll Cardiol 2024;83:2497—604.
8. Zhang H, Jie Y, Wang P, et al. Impact of frailty on all-cause mortality or major amputation in patients with lower extremity peripheral artery disease: a meta-analysis. Ageing Res Rev 2022;79:101656.
9. van Aalst FM, Verwijmeren L, van Dongen EPA, et al. Frailty and functional outcomes after open and endovascular procedures for patients with peripheral arterial disease: a systematic review. J Vasc Surg 2020;71:297—306.e1.
10. Higgins JPT, Thomas J, Chandler J, et al. eds. Cochrane handbook for systematic reviews of interventions. Available at: version 6.4. https://www.training.cochrane.org/handbook. Accessed May 6, 2026.
11. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71.
12. PROSPERO. Internacional Prospective Register of Systematic Reviews. York, (Uk): Centre for Reviews and Dissemination, University of York; National Institute for Health and Care Research (NIHR). Available at: https://www.crd.york.ac.uk/prospero/. Accessed May 6, 2026.
13. Wohlin C. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE'14); 2014 May 13-14; London, UK. New York (NY): Association for Computing Machinery, 2014. pp 1—10. Article 38. Available at: https://doi.org/10.1145/2601248.2601268. Accessed June 5, 2026.
14. Ali TZ, Lehman EB, Aziz F. Modified frailty index can be used to predict adverse outcomes and mortality after lower extremity bypass surgery. Ann Vasc Surg 2018;46:168—77.
15. Balasundaram N, Kanake S, Thaghalli Sunil Kumar V, et al. Modified frailty index as an indicator for outcomes after lower extremity endovascular revascularization. Surgery 2023;173:837—45.
16. Viechtbauer W, Cheung MWL. Outlier and influence diagnostics for meta-analysis. Res Synth Methods 2010;1:112—25.
17. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ 2016;355:i4919.

18. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. BMJ 1997;315:629–34.
19. Andersen JC, Gabel JA, Mannoia KA, et al. 5-Item modified frailty index predicts outcomes after below-knee amputation in the vascular quality initiative amputation registry. Am Surg 2020;86:1225–9.
20. Bradley NA, Walter A, Roxburgh CSD, et al. The relationship between clinical frailty score, CT-derived body composition, systemic inflammation, and survival in patients with chronic limb-threatening ischemia. Ann Vasc Surg 2024;104:18–26.
21. Braet DJ, Taaffe JP, Dombrovskiy VY, et al. Modified frailty index as an indicator for outcomes, discharge status, and readmission after lower extremity bypass surgery for critical limb ischemia. J Vasc Nurs 2020;38:171–5.
22. Brahmbhatt R, Brewster LP, Shafii S, et al. Gender and frailty predict poor outcomes in infrainguinal vascular surgery. J Surg Res 2016;201:156–65.
23. Cotton J, Cabot J, Buckner J, et al. Increased frailty associated with higher long-term mortality after major lower extremity amputation. Ann Vasc Surg 2022;86:295–304.
24. DeRieux J, Obed D, Johnson A, et al. Frailty is not associated with worse outcomes following lower extremity angiograms for limb ischemia in nonagenarians. Ann Vasc Surg 2023;96:261–7.
25. Gonzalez L, Kassem M, Owora AH, et al. Frailty and biomarkers of Frailty predict outcome in veterans after open and endovascular revascularization. J Surg Res 2019;243:539–52.
26. Gonzalez M, Paz M, Babrowski T. Association of frailty index and postoperative outcomes of open bypass lower extremity revascularization for acute limb ischemia using the vascular quality initiative. Vasc Endovascular Surg 2025;59:387–95.
27. Hart JP, Davies MG. Transitions of frailty after lower extremity interventions for chronic limb-threatening ischemia. J Vasc Surg 2025;81:730–742.e4.
28. Houghton JS, Nickinson AT, Helm JR, et al. Associations of clinical frailty with severity of limb threat and outcomes in chronic limb-threatening ischaemia. Ann Vasc Surg 2021;76:406–16.
29. Houghton JSM, Black I, Meffen A, et al. Relationship of the wound, ischemia, and foot infection stage with Frailty, disability, quality of life, and one-year outcomes in all-comers with chronic limb-threatening ischemia: single-center prospective cohort study. Ann Vasc Surg 2026;124:109–19.
30. Karim AM, Li J, Panhwar MS, et al. Impact of malnutrition and frailty on mortality and major amputation in patients with CLTI. Catheter Cardiovasc Interv 2022;99:1300–9.
31. Kodama A, Koyama A, Sugimoto M, et al. Association between preoperative frailty and mortality in patients with critical limb ischemia following infrainguinal bypass surgery—usefulness of the Barthel Index. Circ J 2017;82:267–74.
32. Majmundar M, Wan-Chi C, Patel KN, et al. Prognostic value of the Hospital Frailty Risk Score (HFRS) and outcomes in peripheral artery disease and end-stage kidney disease. Vasc Med 2025;30:138–46.
33. Pandit V, Tan TW, Kempe K, et al. Frailty syndrome in patients with lower extremity amputation: simplifying how we calculate frailty. J Surg Res 2021;263:230–5.
34. Schweitzer CD, Anagnostakos JP, Nagarsheth KH. Frailty as a predictor of adverse outcomes after peripheral vascular surgery in patients with end-stage renal disease. Am Surg 2022;88:686–91.
35. Soon SX, D'Çruz R, Yap CJ, et al. The modified frailty index-11 predicts medium-term outcomes after endovascular revascularisation for chronic limb threatening ischaemia in Asian patients. Vascular 2022;30:42–51.
36. Butala NM, Raja A, Xu J, et al. Association of frailty with treatment selection and long-term outcomes among patients with chronic limb-threatening ischemia. J Am Heart Assoc 2021;10:e023138.
37. Damluji AA, Chung SE, Xue QL, et al. Frailty and cardiovascular outcomes in the national health and aging trends study. Eur Heart J 2021;42:3856–65.
38. Zhao Y, Wu Y, Liu Z, et al. Association of frailty and pre-frailty with cardiovascular mortality: a meta-analysis of 26 cohort studies. Front Public Health 2025;13:1688014.
39. Shrauner W, Lord EM, Nguyen XMT, et al. Frailty and cardiovascular mortality in more than 3 million US Veterans. Eur Heart J 2022;43:818–26.
40. Welsh SA, Pearson RC, Hussey K, et al. A systematic review of frailty assessment tools used in vascular surgery research. J Vasc Surg 2023;78:1567–1579.e14.
41. Majmundar M, Patel KN, Doshi R, et al. Prognostic value of hospital frailty risk score and clinical outcomes in patients undergoing revascularization for critical limb-threatening ischemia. J Am Heart Assoc 2023;12:e030294.